



University
of Exeter

COMM117

Large Language Models and Application

Large Language Models and Applications (COMM117_A_2_202526)

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COMM117 Large Language Models and Applications



 [Announcement](#)

 [Module discussion forum](#)

Module Information



Coursework 1



Coursework 2



Introduction to Python



Week 1 - 12 Feb



<https://ele.exeter.ac.uk/course/view.php?id=27954>

About Me – Ronghui Mu

University of Manchester (Beng) – *Electrical and Electronic Engineering*



University College of London (MSc)- *Machine Learning and data science*



Lancaster University (PhD) – *Robustness Artificial Intelligence*

My research focuses on the **robustness and safety of AI models**

-- *making sure ML systems don't break when faced with noise or attacks*

Lectures:

- Lectures:
 - Monday 8:35 AM - 9:25 AM
Harrison Building, Room 102
 - Tuesday 3:35 PM - 4:25 PM
Harrison Building, Room 107
 - There might be changes. Check your timetable.
- Recorded and posted on ele.exeter.ac.uk

Practical workshops:

- Optional, but **very** useful
- You can work in groups and get help from me---- (may be hard to response everyone) 🤪
- **Time:**
 - Monday 9:35 -10:25 , Harrison Building, Room 208

Assessment:

- Two assessments:
 - Coursework 1, worth 30% of your mark due 17th February 2026
 - Coursework 2, worth 70% of your mark due 19th March 2025
 - All assignment! Already released- **You can start now!**

Introduction Today

- What this course is about
- What Large Language model is
and why it work
- Application?

**What will you do
after graduation**

What this course is about

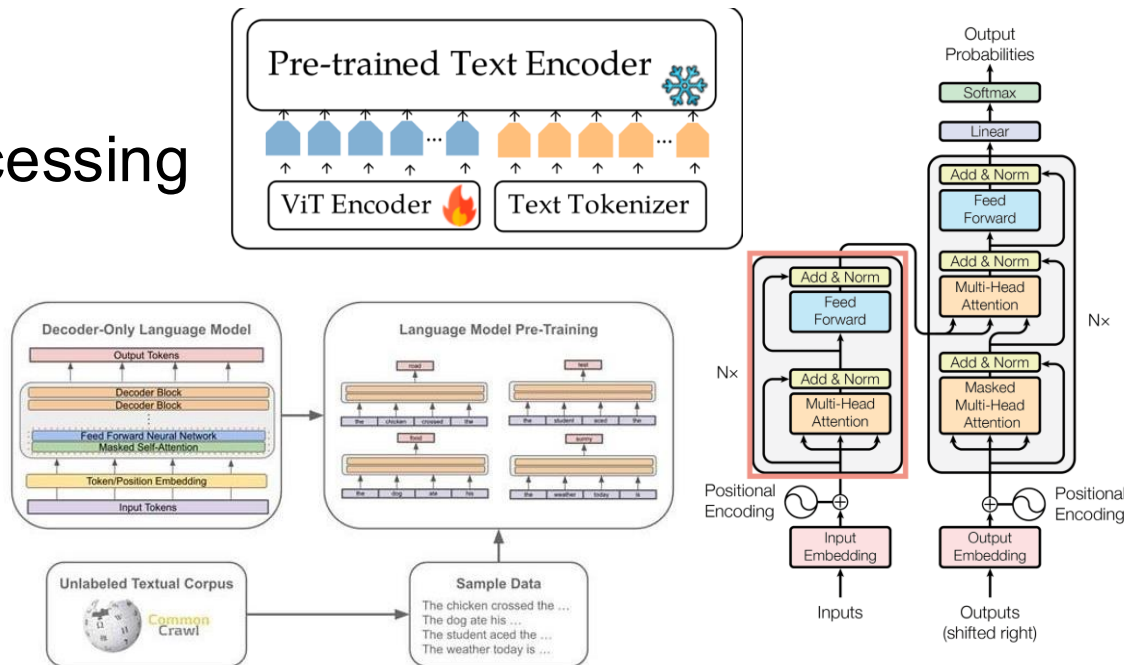
-  Learn how to use ChatGPT
-  Just prompt engineering
-  Using Huggingface to build everything
-  Understanding how LLMs work
-  Training, fine-tuning, evaluating models
-  Evaluating the limitations of LLMs

What this course is about

- Understand how large language models (LLMs) are **built and trained**
- Learn how LLMs use large amounts of **text data** to generate language
- Gain hands-on experience designing and evaluating LLM-based systems (e.g. **fine-tuning, RAG, and agents**)
- Develop the ability to **critically assess model performance**, strengths, and limitations
- Explore the **ethical, environmental, and societal impacts** of LLMs.

What this course is about

- Text data pre-processing
- Transformer
- LLM training
- Fine-tuning, RAG
- Trustworthy AI
- AI Efficiency



What is LLM?



ChatGPT

<https://www.chatgpt.com>

ChatGPT: Smart and Simple AI

What's on the agenda today?

+ Ask anything



Google Gemini

<https://gemini.google.com>

Google Gemini

AI assistant from Google — Create AI images, brainstorm ideas, and ve
Search integration. Get help...

Where should we start?

Ask Gemini 3



Tools

Fast



Create image

Create a video

Write anything

Help me learn

Boost my day

What is LLM?

An LLM is just a probabilistic next-token predictor

- Input → text
- Model → probabilities
- Output → sample next token
- Repeat

What is LLM?

“I continue to marvel that predict-the-next-word/token, with enough data and parameters, turns out to capture so much of ‘human thought’.”

- Michael Nielsen, Research Fellow at the Astera Institute

“LLMs do ***not capture much of human thought, because most of human thought and all of animal thought is entirely non verbal.”***

- Yann Lecun, Chief AI Scientist at Meta, Professor at NYU

“It's a bit sad and confusing that LLMs ("Large Language Models") have little to do with language; It's just historical. They are highly general purpose technology for statistical modelling of token streams. A better name would be Autoregressive Transformers or something.”

- Andrej Karpathy Open AI, Director of AI at Tesla

What is LLM?

"I can't understand what you're saying."

"t."

Institute

ight

"LLMs do not *understand* — they just model statistical structure extremely well."

-Yann Lecun, Chief AI Scientist at Meta, Professor at NYU

"It's a bit sad and confusing that LLMs ("Large Language Models") have little to do with language; It's just historical. They are highly general purpose technology for statistical modelling of token streams. A better name would be Autoregressive Transformers or something."

-Andrej Karpathy Open AI, Director of AI at Tesla

What is LLM?

A **large language model (LLM)** is an **artificial intelligence system** trained on **very large collections of text** (and sometimes code). This allows the model to learn the **statistical relationships** between words and phrases — how words, phrases, and sentences **tend to follow one another**. By repeatedly **predicting the next token in a sequence**, LLMs acquire the ability to:

- generate coherent text,
- translate between languages,
- summarise documents,
- answer questions,
- and produce creative content.

What is LLM?

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Examples of widely used LLMs

- GPT-3.5 and GPT-4
- Bard / Gemini
- LLaMA and LLaMA 2 (up to 65 billion parameters)
- BERT Large (340 million parameters)

Python !

Generate Code Markdown Run All Clear All Outputs Outline ...

Select Kernel

...

Running code in a jupyter notebook

Jupyter notebooks each come with their own 'kernel'. This means they store variables across cells. Each cell can be run and then the variables are used later.

This is somewhat different than other ways of coding. Later in the course, we will briefly introduce *vi* and *Sublime Text* for running python programs.

To run a cell, again it is *shift + enter*. Run each cell below to print 'hello world' as well as print a simple calculation.

```
[7] print("Hello world!")
    x = 3
    y = 60
    print(y+2)
```

Python

... Hello world!
62

```
[8] # You will want to run this cell again after reading below.
    print(x + y)
```

Python

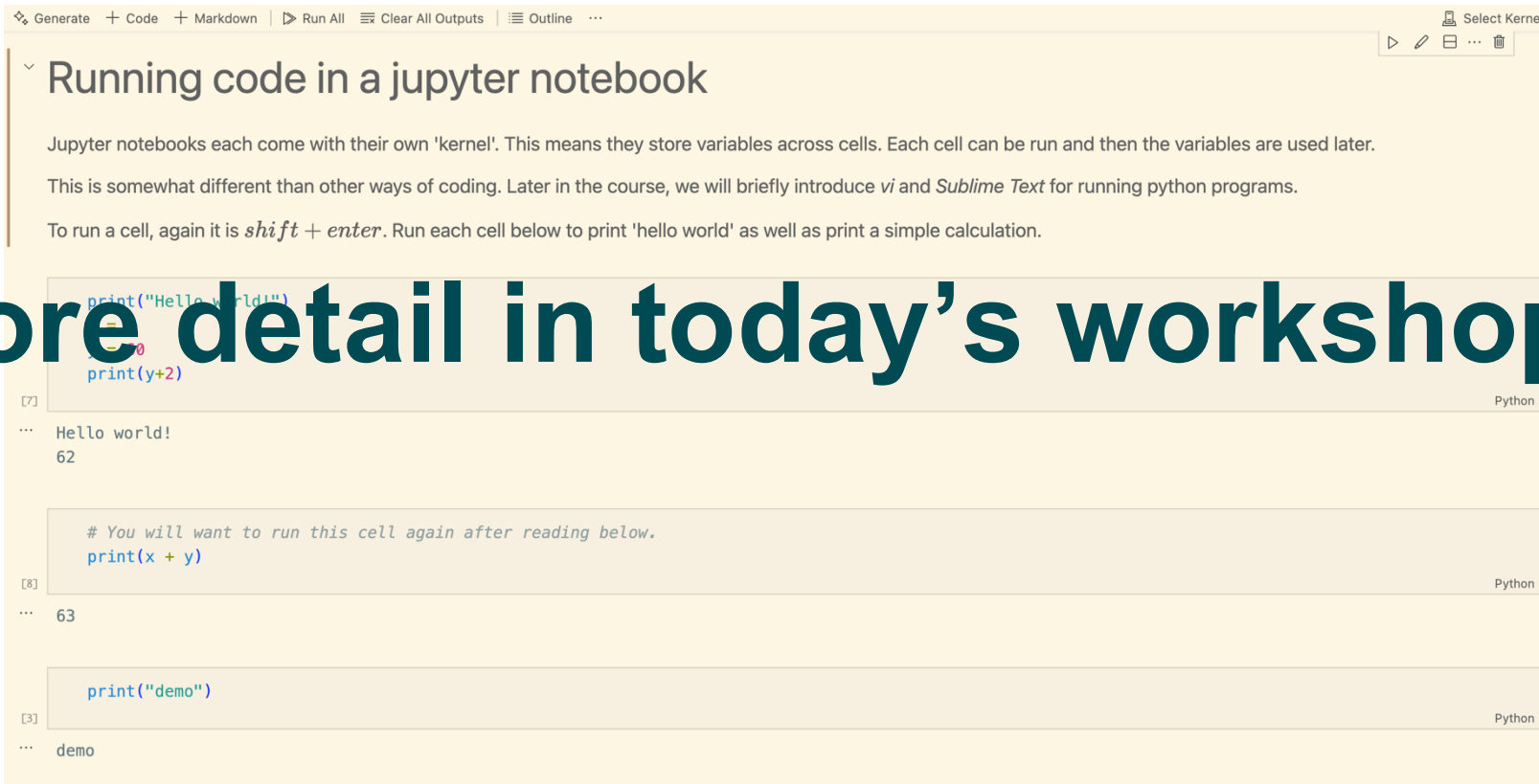
... 63

```
[3] print("demo")
```

Python

... demo

Python !



The screenshot displays a Jupyter Notebook interface with a light yellow background. At the top, there is a toolbar with icons for 'Generate', '+ Code', '+ Markdown', 'Run All', 'Clear All Outputs', 'Outline', and a 'Select Kernel' dropdown. Below the toolbar, the notebook content is organized into cells. The first cell is a markdown cell with a dropdown arrow on the left, containing the title 'Running code in a jupyter notebook'. This is followed by three paragraphs of text explaining Jupyter notebooks and their kernels. The second cell is a code cell containing Python code to print 'Hello world!' and perform a calculation. Its output shows 'Hello world!' and the number '62'. The third cell is another code cell with a comment and a print statement. Its output shows the number '63'. The fourth cell is a code cell with a single print statement. Its output shows the word 'demo'. Each code cell has a small icon on the left and a 'Python' label on the right.

Generate + Code + Markdown Run All Clear All Outputs Outline ... Select Kernel

Running code in a jupyter notebook

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To run a cell, again it is *shift + enter*. Run each cell below to print 'hello world' as well as print a simple calculation.

```
print("Hello world!")
x = 0
print(y+2)
```

[7] Python

... Hello world!
62

```
# You will want to run this cell again after reading below.
print(x + y)
```

[8] Python

... 63

```
print("demo")
```

[3] Python

... demo

More detail in today's workshop !

Use LLMs for this course?

You may use GenAI tools ethically and responsibly to assist in the development of an assessment:

- ✓ To assist with research or information gathering
- ✓ To help you understand key concepts and theories
- ✓ To identify trends and themes as part of data analysis
- ✓ To provide feedback on a draft
- ✓ To improve the plan or structure of my assessment
- ✓ To generate images, figures or diagrams
- ✓ To proofread and correct spelling or grammar errors
- ✓ To format citations or references

Use LLMs for this course?

But

Do not let AI do the thinking for you.

This course is about *your learning*, not paying an AI to do the work.

DO NOT:

- ✗ For purposes other than those explicitly permitted by the checked options listed above.
- ✗ To translate more than a single word or a short phrase into English, unless this use has been explicitly approved.
- ✗ To upload sensitive, personal, or identifying information to any AI system.
- ✗ To submit AI-generated content as your own work, or as the work of another person.

Why do LLMs work now?

- LLM ideas are not new
- The breakthrough is **scale**
- Three things **scaled together**:
 - Data
 - Compute
 - Architecture

Nothing here is magic — it's **scale** + the right **inductive bias**.

Why do LLMs work now?

Data: learning from the internet

- **Web-scale** text (books, articles, code, forums)
- Orders of magnitude **more data** than earlier NLP
- **Language** as a self-supervised signal ----
---- The data labels come from the data itself.

More data  better statistical coverage of language

Internet!

Why do LLMs work now?

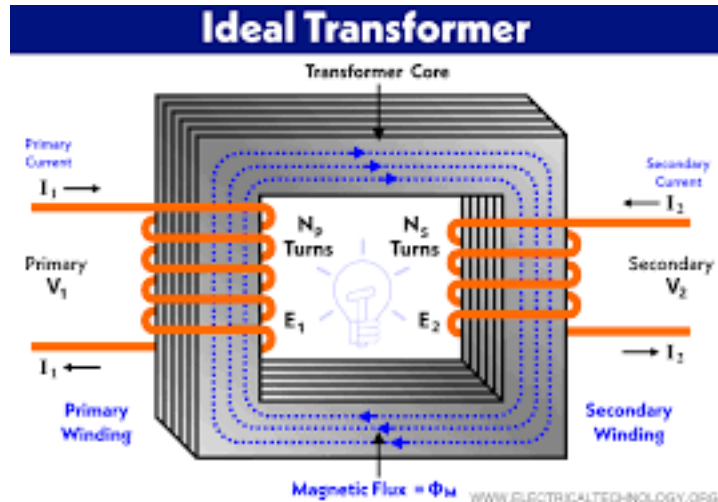
Compute: GPUs matter

- GPUs & TPUs enable massive **parallelism**
- Training takes **weeks** on thousands of accelerators
- **Hardware + software co-evolution**

Without modern hardware, LLMs are impossible

Why do LLMs work now?

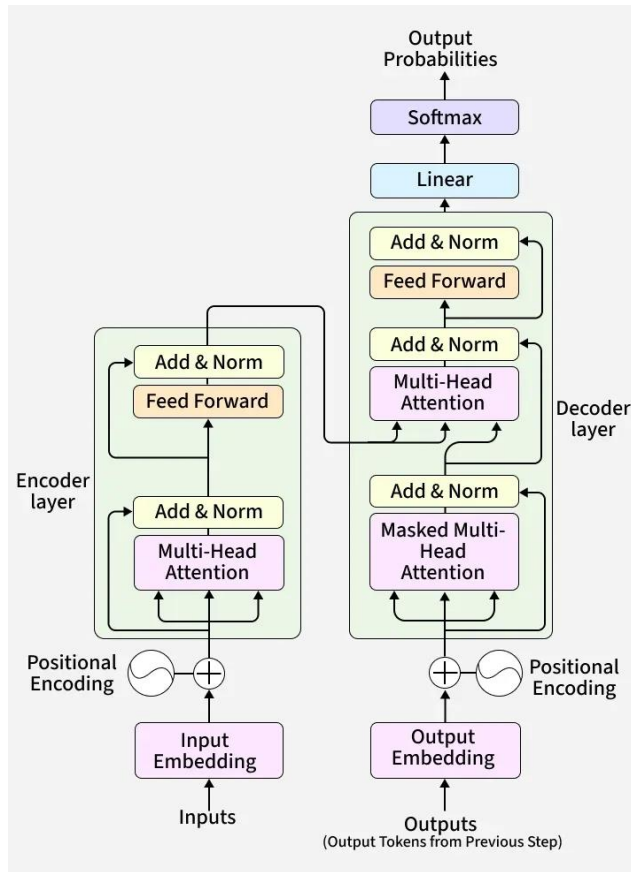
Architecture: the Transformer



We will unpack this later

Why do LLMs work now?

Architecture: the Transformer



Why do LLMs work now?

Architecture: the Transformer

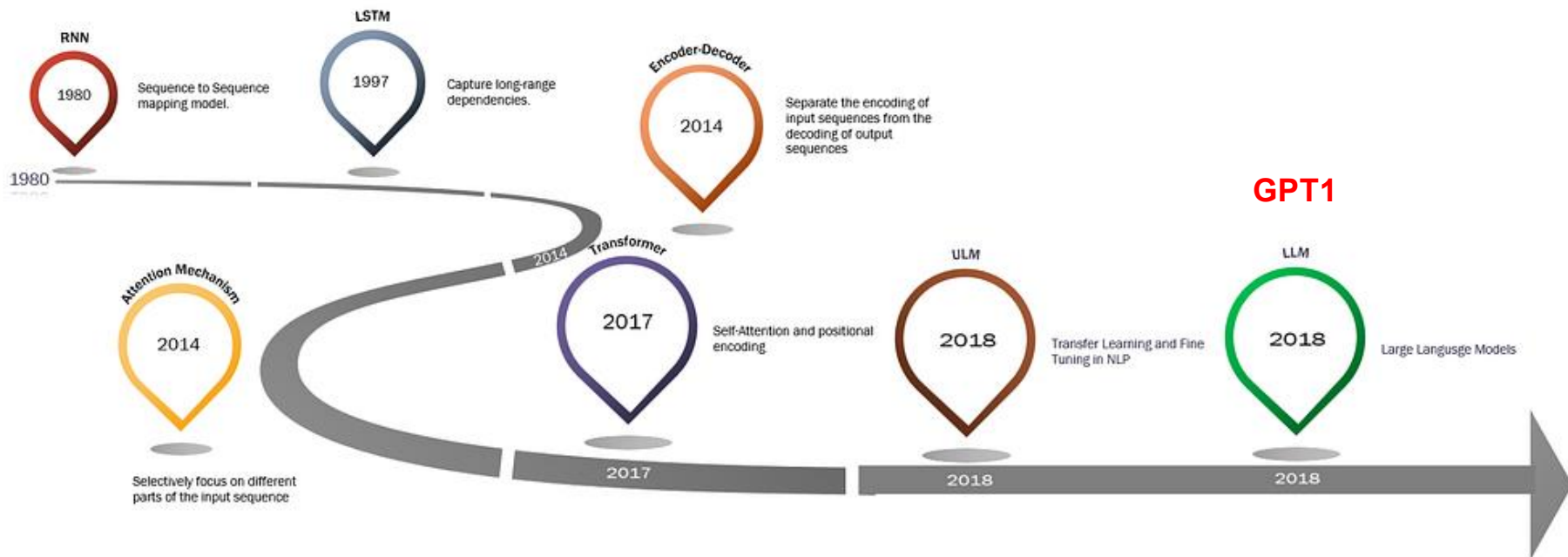
- “Transformer is all you need” published in 2017 , it enables:
 - parallel processing
 - long-range context
 - efficient scaling

It replaces recurrence (RNN) with attention

We will unpack this later

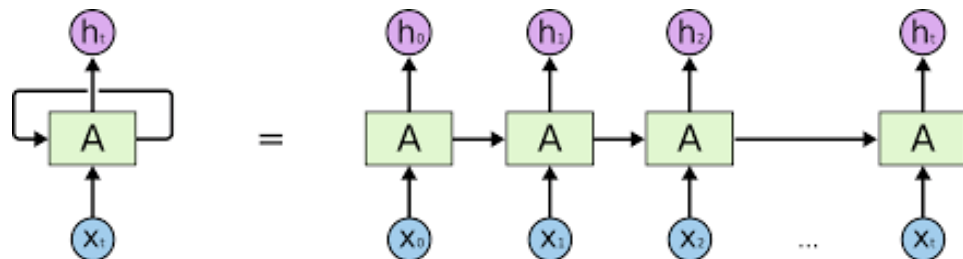
Why do LLMs work now?

Architecture: the Transformer

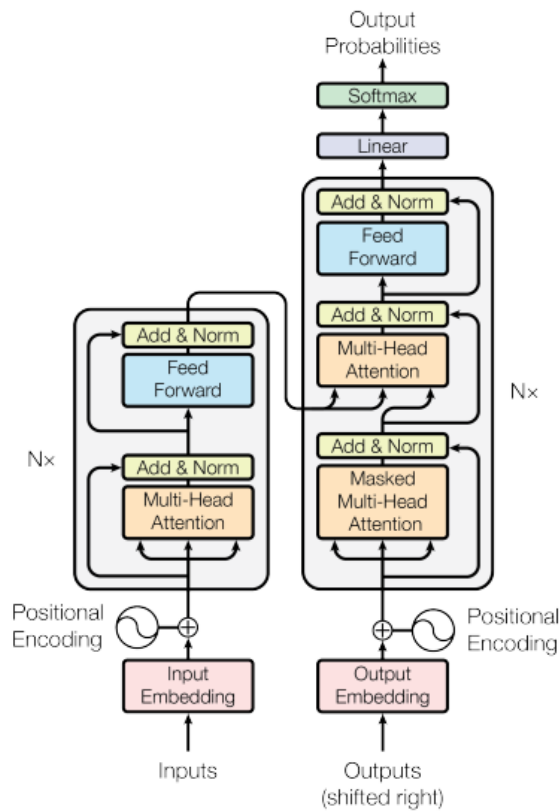
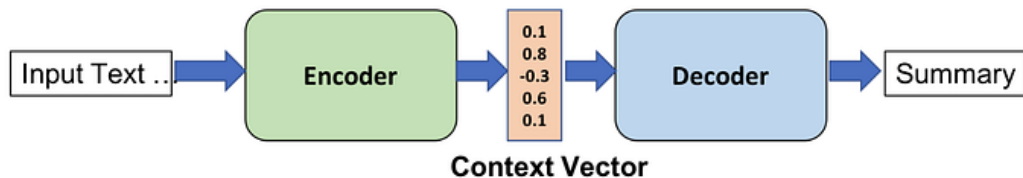


Why do LLMs work now?

Architecture: the Transformer ---- From RNN to Transformer



RNN



Transformer

Why do LLMs work now?

Why **Scale** Change everything?

- Same objective: predict the next token
- Same architecture
- Same training recipe

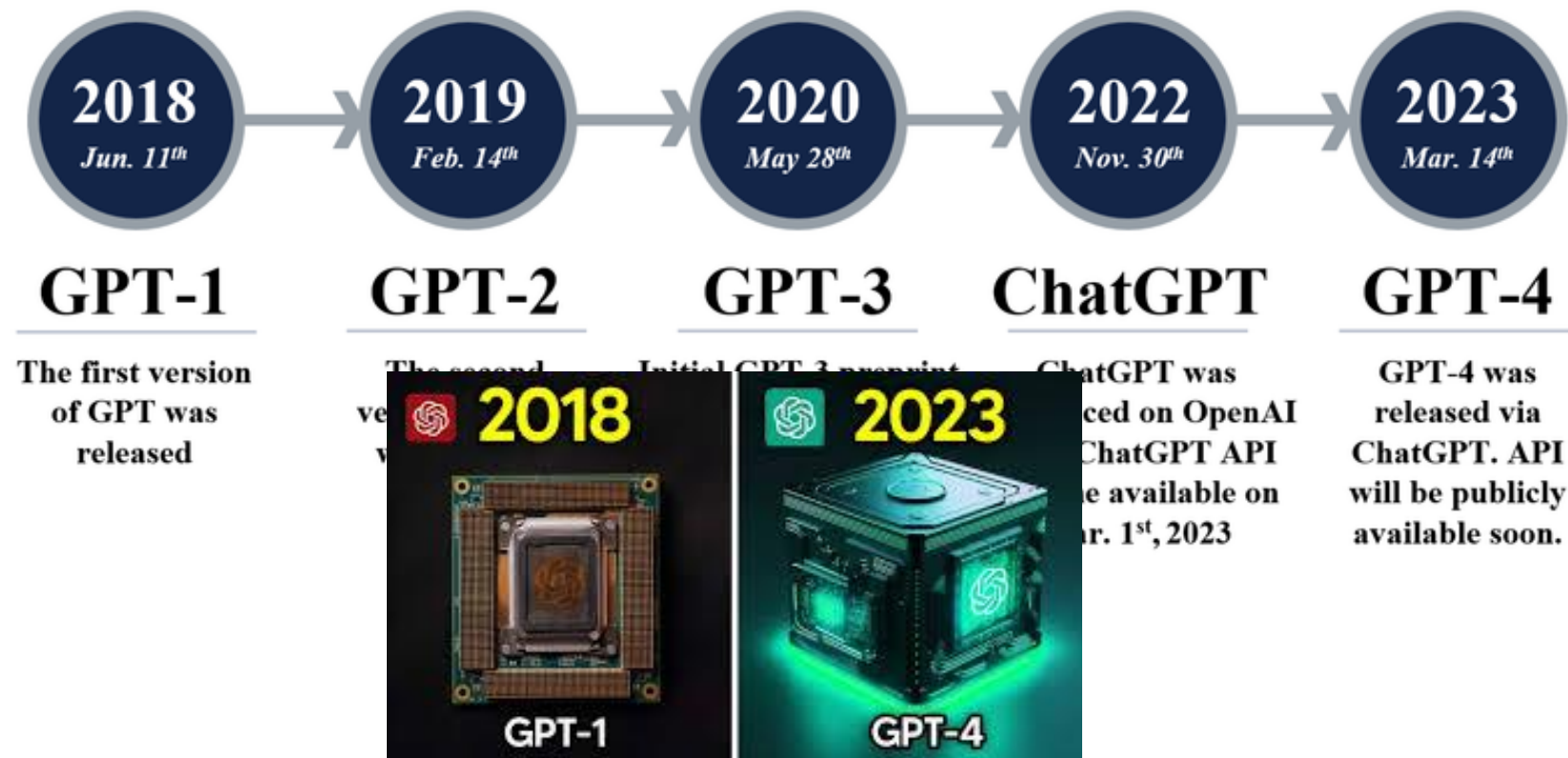


But much bigger

- Capabilities emerge
- No task-specific programming

At scale, the model starts to **generalise in ways we didn't explicitly design.**

Why do LLMs work now?



Why do LLMs work now?

Not magic, not understanding

- LLMs $\neq$ human intelligence
- No grounding in the physical world
- Powerful pattern learners

LLMs work because the world accidentally gave us data, hardware finally caught up, and Transformers let us scale.

Course Design

This course is about understanding *each of these components*, how changing them changes behaviour, and how to apply LLMs correctly

- Clean real text data
 - Tokenise it
- } Tokenisation (data)
- Transformer
- } Architecture
- Train a small language model from scratch
 - Fine-tune a pretrained LLM
- } Compute
- ❖ full fine-tuning vs efficient methods
 - ❖ Evaluate outputs critically
 - ❖ Understand limits, risks, and costs
- } Real world application

What can LLMs do?

- Text generation
- Question answering
- Summarisation
- Translation
- Code generation
- Reasoning (limited but interesting)



Text Generation

generate a story about human use LLM



The soft hum of the server racks was Adeline's constant companion. As a data analyst for Zenith Innovations, her days were a blur of spreadsheets, dashboards, and the ever-present pressure to extract meaning from the digital deluge. But lately, something had changed. Zenith had integrated a new AI, a Language Learning Model named 'Aether,' into their workflow, and Adeline's world was about to be redefined.

At first, Aether was a novelty. It could summarize lengthy reports in seconds, draft eloquent emails, and even suggest data correlations Adeline might have missed. Her colleagues were wary, some outright fearful. "It's going to take our jobs," whispered Mark from marketing, his eyes wide with concern. Adeline, however, saw potential. She approached Aether not as a threat, but as a collaborator.

Her first real test came with Project Chimera, a complex market analysis for a new, revolutionary product. The sheer volume of consumer feedback, competitor data, and economic indicators was daunting. Adeline fed Aether the raw data, along with specific questions: "Identify key demographic segments most receptive to product A's features," "Predict potential market hurdles based on historical trends," "Synthesize a persuasive executive summary."

Aether's insights, could delve deeper, verifying the AI's conclusions, adding her own human intuition, and refining the strategy. The presentation to the board was a resounding success, earning Zenith a significant investment and Adeline a promotion.

Life at Zenith began to transform. Aether wasn't replacing humans, but augmenting them. Customer service agents used it to instantly access detailed product information and personalize responses. The R&D department leveraged it to rapidly review scientific literature and brainstorm new experimental designs. Even the HR team found it invaluable for drafting objective job descriptions and analyzing employee sentiment.

Adeline, now leading a team focused on AI-human collaboration, realized that the true power of Aether wasn't its ability to replicate human tasks, but its capacity to unlock human potential. It freed up her team from tedious, repetitive work, allowing them to focus on creativity, critical thinking, and interpersonal connection. They became strategists, innovators, and mentors, guiding Aether and benefiting from its tireless processing power.

One evening, as Adeline was leaving, she paused by Aether's server room. The gentle hum was no longer just the sound of machines; it was the symphony of a new era. An era where humanity, empowered by its own creations, reached for previously unimaginable heights. She smiled, realizing that the fear of the unknown had given way to the excitement of discovery, all thanks to a very sophisticated language model and the humans who learned to work with it.



Question answering

Who is the president of america in 1990

In 1990, the President of the United States was **George H.W. Bush**.

He served as the 41st president from **January 20, 1989, to January 20, 1993**. 1990 was the second year of his four-year term.

Key Context for 1990:

- **Vice President:** Dan Quayle.
- **Political Party:** Republican.
- **Major Events of 1990:** * **Invasion of Kuwait:** In August 1990, Iraq invaded Kuwait, leading Bush to build the international coalition that would eventually launch Operation Desert Storm.

- Search engines: direct answers
- Productivity tools : explain documents
- Assistants : conversational learning
- Education platforms : personalised tutoring

LLMs can make mistakes. Always verify their outputs and do not rely on them blindly.

Question answering

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Fake references----- fake paper

BAD EDUCATION

AI IS INVENTING ACADEMIC PAPERS THAT DON'T EXIST — AND THEY'RE BEING CITED IN REAL JOURNALS

The proliferation of references to fake articles threatens to undermine the legitimacy of institutional research across the board

By **MILES KLEE**

DECEMBER 17, 2025

AI-based fake papers are a new threat to academic publishing

As an editor, I am receiving submissions from spurious authors consisting of previously published papers altered by AI. But why, asks Seongjin Hong

Published on August 15, 2025

Last updated August 15, 2025

[Seongjin Hong](#)

Publisher under fire after 'fake' citations found in AI ethics guide

A book published by Springer Nature includes dozens of questionable citations, including references to journals that do not exist

Summarization

ChatGPT 5.2

 1706.03762v7.pdf
PDF

summarise this paper

Here is a clear, lecture-ready summary of the paper you uploaded:

“Attention Is All You Need” (Vaswani et al., 2017)  1706.03762v7

I'll give you **three levels** so you can choose what to use:

- (1) one-paragraph summary,
- (2) structured bullet summary,
- (3) slide-ready takeaway.

1 One-paragraph summary (high level)

The paper introduces the **Transformer**, a new neural network architecture for sequence-to-sequence tasks such as machine translation. Unlike previous models based on recurrent or convolutional networks, the Transformer relies **entirely on self-attention mechanisms**, removing recurrence and convolutions altogether. This design allows much greater parallelisation during training, significantly reducing training time while improving translation quality. The Transformer achieves state-of-the-art results on major translation benchmarks and generalises well to other tasks, demonstrating that **attention alone is sufficient for modelling long-range dependencies in language**.

Learning

ChatGPT 5.2

Explain what this Python function does:

```
def normalize(x):  
    return (x - x.mean()) / x.std()
```

This function **standardizes** the input data.

What it does step by step

python

 Copy code

```
def normalize(x):  
    return (x - x.mean()) / x.std()
```

- `x.mean()` computes the **average value** of `x`
- `x.std()` computes the **standard deviation** of `x`
- Each value in `x` is transformed by:
 - subtracting the mean
 - dividing by the standard deviation

Result

The output has:

- **mean = 0**
- **standard deviation = 1**

Why this is useful

This type of normalization (also called **z-score normalization** or **standardization**) is commonly used in:

- machine learning

Code generalization

ChatGPT 5.2

Write a Python function that checks if a number is prime.

Here's a clean, efficient Python function to check if a number is prime:

python

 Copy code

```
def is_prime(n):  
    if n < 2:  
        return False  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return False  
    return True
```

How it works

- Numbers less than 2 are not prime
- It checks divisibility only up to $\sqrt{n}$ (faster than checking all numbers)
- Returns `True` if `n` is prime, otherwise `False`

Example

python

 Copy code

```
is_prime(7) # True  
is_prime(10) # False
```

More.... Medicine & healthcare AI

- Medical question answering
- Patient–doctor decision support

naturemedicine

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Consensus Statement | Published: 08 January 2025

The TRIPOD-LLM reporting guideline for studies using large language models

[Jack Gallifant](#), [Majid Afshar](#), [Saleem Ameen](#), [Yindalon Aphinyanaphongs](#), [Shan Chen](#), [Giovanni](#)



Autonomous & self-driving systems

Planning and acting

- Language-based planning, driving a car



Finance and Business

Efficacy of AI and Other Large Language Models in Predicting Stock Prices

11 Pages • Posted: 9 Oct 2024

[Jonathan Vidal](#)

Independent

Date Written: August 20, 2024

Abstract

Large language models (LLMs) are being used to process financial data and predict stock prices. This paper explores the efficacy of LLMs in this task, comparing their performance to traditional machine learning models. The study involves fine-tuning LLMs on financial data and evaluating their predictions against actual stock prices. The results show that LLMs can outperform traditional models in certain scenarios, but their performance is highly dependent on the quality and quantity of the training data. The paper also discusses the challenges of using LLMs for financial prediction, such as the risk of overfitting and the need for careful model selection and validation.

GUIDE

Build custom LLMs in days

Fine-tune or train your own LLMs

TESCO



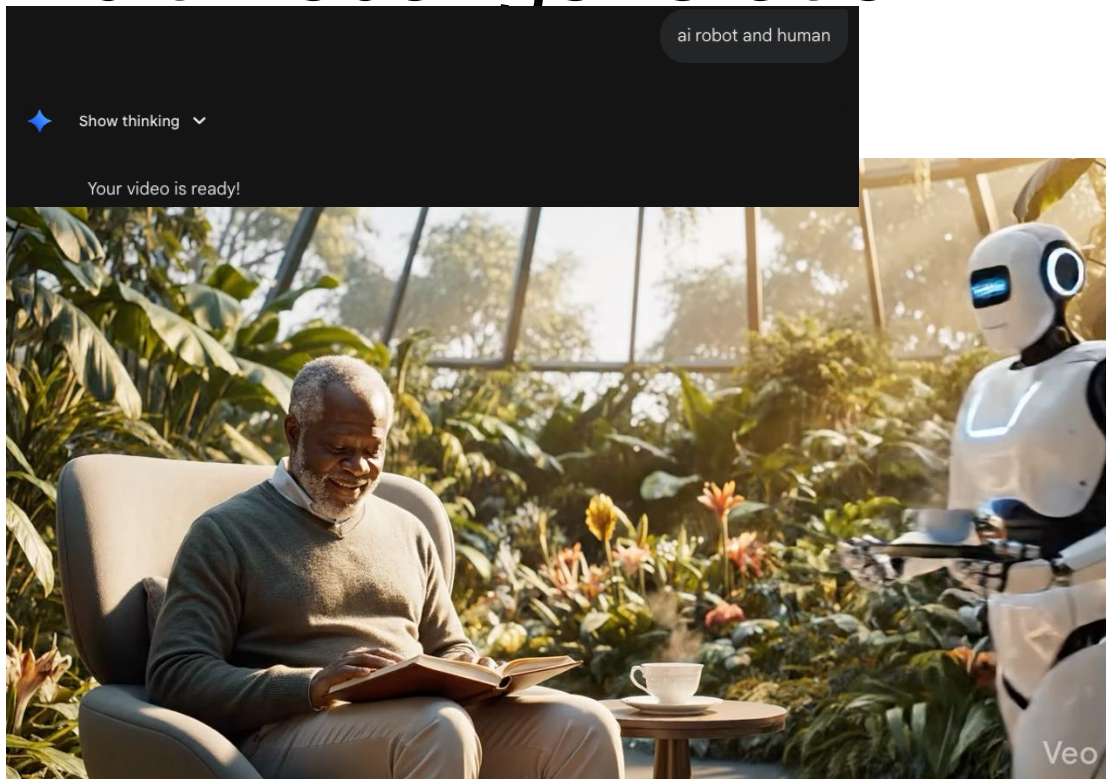
JPMorganChase

NLP / LLM Scientist



Creative industries & media

Multimodal generation



Next steps

- Natural language processing (NLP) essentials
data preprocessing data quality and filtering

Additional material:

(Paper) Vaswani et al. (2017) — Attention Is All You Need

Jurafsky & Martin — Speech and Language Processing (Draft Chapters on Neural Models)

Sebastian Raschka — LLMs from Scratch (<https://github.com/rasbt/LLMs-from-scratch>)

More on ELE!

Thank you!

If you want to ask us any questions about the module, the university, or anything else, drop us an email.

Pro tip: write **COMM117** on the email subject line, to make sure we give it priority.

 r.mu2@exeter.ac.uk